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Improving the typing of *Xylella fastidiosa* directly from plant material

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Abstract: Detection of *X. fastidiosa* directly from plant material is a common practice as a screening test for epidemiosurveillance purposes. MultiLocus Sequence Analysis and Typing (MLSA/MLST) methods are amongst the most widely used genotyping methods for assessing the global epidemiology of various plant pathogenic bacteria including *X. fastidiosa*. MLSA is used to assign strains to one of the known *X. fastidiosa* subspecies and MLST for typing strains and make inference of their origin. Both methods are traditionally used on isolated strains. Nonetheless, because it is time-consuming and sometimes difficult to obtain isolates of *X. fastidiosa* from plant samples as a consequence of an intrinsically poor isolation rate and the fastidious nature of the pathogen, being able to type *X. fastidiosa* in plant samples is a priority for epidemiological purposes. Application of the initial protocol on DNA extracted from plant material proved limited efficiency. The aim of our study was to test the impact of various modifications of the protocol in order to improve the efficiencies of DNA extraction and target amplification rate. We also tested the interest of reduced typing schemes. The direct identification of *X. fastidiosa* infecting plant material face however some limitations that will be discussed.

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Real-time LAMP rapid diagnostic method for *X. fastidiosa* in plant material and insect vectors

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Abstract: *X. fastidiosa* is a gram-negative insect-vectored bacterium has been recently detected in Italian olive trees severely affected by Olive Quick Decline Syndrome (OQDS). ELISA tests and Polymerase Chain Reaction (PCR) assays were largely used in the monitoring campaign of this pathogen. The aim of this study is to evaluate the new real time Loop-mediated isothermal amplification (LAMP) system for the detection of *X. fastidiosa* in host plants and insect vectors. The new detection system is composed of a portable instrument (icgene mini) and a ready to use diagnostic kit denominated "Xylella Screen Glow" (Enbiotech s.r.l.- Italy). Specificity and sensitivity of Xylella Screen Glow kit was compared with PCR and real-time quantitative PCR assays. Three different DNA extraction protocols and typologies of infected materials were also tested. The real-time LAMP system showed high specificity and a sensitivity as compared to the real-time qPCR and PCR assays. These results showed that "Xylella Screen Glow" kit with icgene portable LAMP instrument is highly